NORTHERN WOLVES
AIR CONDITIONING

Request for Information

Project	Seafarers International House - 18 Vesey St, New York, NY 10007 (KBD job 30-26004)
General Contractor	KBD Group (Kajima Building & Design Group, Inc.) - Janina Leshik, Peter Sawiris
Engineer	All City Engineering, P.C. (ACE)
Architect	Arcari + Iovino Architects PC
RFI #	NWAC 04, Rev. 0
Date	10/30/2026
Subject	Fuel oil supply / return (Brugg FLEXWELL-HL) - leak detection probe cable and second 2-hr enclosure (ACE comments on submittal 230000-19.2)
Drawing No.	FO-1 (submittal 230000-19.2); FS101, FS300; M-101.00

Contents

1. Question and proposed solution (page 2).
Attachment A: Brugg FLEXWELL-HL data sheet, UL 1369 lines highlighted (pages 3-4).

Reference documents

Document	Status
Submittal 230000-19.2, FO-1 Fuel Oil Piping Layout (Rev. 02, 9/17/2026)	Make Corrections Noted 10/20/2026
Submittal 230000-15.2, Core Engineered Solutions SBM5622 R002 (tank, pump set, controls, Brugg piping, vacuum leak monitor)	No Exceptions Taken 6/11/2026
Brugg FLEXWELL-HL data sheet (UL 1369 "Severe", above ground)	Attachment A
FS101-FS105, FS300, M-101.00 (Rev 0, issued for bid 1/8/2026)	Brugg double-wall piping with vacuum leak panel, in a single 2-hr rated sheetrock enclosure

Submitted by / distribution

Submitted by: Alikhan Ilyas, Northern Wolves AC - alikhan@northernwolvesac.com - (347) 463-9248
Distribution: KBD Group (Janina Leshik, Peter Sawiris); Arcari + Iovino (Chris Santos); ACE.

1. Question

ACE's review of submittal 230000-19.2 (10/20/2026) added two requirements to the 1" FOS / 1-1/2" FOR Brugg FLEXWELL-HL piping from the sub-cellar to the roof generator:

#	ACE comment	Contract documents
A	"Oil leak detection probe cable to be provided." (no location, length or product given)	Not shown on any sheet. FS101 / FS300 show the "BRUGG VACUUM SYSTEM" / "BRUGG VACUUM LEAK PANEL" only.
B	"Supply and return piping shall be enclosed in another layer of 2 hr rated enclosure." / "The pipe enclosure shall have another 2 hr rated enclosure."	One enclosure: FS300 "FUEL OIL RISER SHALL BE 2 HR RATED SHEETROCK ENCLOSURE"; plans: "BRUGG PIPING IN 2 HR RATED SHEETROCK ENCLOSURE".

Brugg FLEXWELL-HL listing (Attachment A)

Data sheet	Bearing on the comment
"FLEXWELL-HL has a two hour fire rating per UL 1369."	B: the pipe itself carries a 2-hr rating, inside the 2-hr enclosure already on the drawings.
"No requirement for constant leak monitoring per UL 1369 for BRUGG FLEXWELL-HL."	A: the listing does not require monitoring. The project still adds continuous vacuum monitoring (Core FOC-LMS-VAC-4) of the full interstice, with leak sensors at the tank basin and pump skid.
Continuous lengths, no intermediate joints; closed mechanical end fittings	A: the only points where oil can leave the double wall are the two end fittings.

Proposed solution

A. No probe cable. Keep the approved vacuum monitoring and add one Core 00L-LS point leak sensor at each Brugg end-fitting location (sub-cellar at pump set FOP-1, roof at the generator tank connection), wired to the FOC panel and annunciator.
B. Single 2-hr rated enclosure as shown on the contract drawings. No second enclosure layer.

Requested from the Engineer

1. Confirm items A and B as proposed.
2. If a probe cable or a second enclosure layer is still required, define the locations, extent and product or construction. Neither is shown on the contract documents; the added work, including any re-routing to keep the required 3 ft x 7 ft clearance, will be priced as a change to the contract.

BRUGG
Pipes

UL 1369 "Severe" Certified

FLEXWELL®-HL

Double Wall Stainless Steel Pipe for Above Ground, Below Ground, Marina and Commercial Fuel Systems

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Flexible, double-wall corrugated pipe system with continuous interstice for leak detection

GENERAL DESCRIPTION

The FLEXWELL®-HL pipe system is specifically designed for hazardous fluids and gases, such as gasoline, ethanol blends, biodiesel, LPG and other hazardous fluids typically transported in chemical plants. The pipe is used in gas stations underground as well in above ground applications such as marinas and bulk facilities. FLEXWELL®-HL can operated as a suction or pressure systems and can be continuously monitored via suitable and approved leak detection systems.

CONSTRUCTION

FLEXWELL®-HL is a flexible pipe system made of corrugated 316lL (1.4404) stainless steel primary and secondary pipes for operating pressures of up to 145 PSIG (10 bar). The interstital space between primary and secondary pipe can be used for leak detection. The pipe is corrosion protected by a polyethylene jacket. The stainless steel pipe provides an enduring, impermeable barrier, even with future new fuels or fuel combinations.

APPLICATIONS

- suction pipes
- pressure pipes
- fill pipes

SIZES AND PRESSURE RATINGS

FLEXWELL®-HL is available with an ID of 1" (ND 25) to 3" (ND 80) for conventional use at an operating pressure of up to 145 PSIG (10 bar) and temperatures between – 50 °C (– 58 °F) to + 60 °C (+ 140 °F).

INSTALLATION

FLEXWELL®-HL is manufactured in factory lengths up to 3,000 ft. FLEXWELL®-HL can be laid directly into a trench in one piece following any directions without necessity of intermediate joints or fittings. The corrugated design of the primary and secondary pipes assure excellent impact resistance while at the same time providing a high degree of flexibility.

TWO HOUR FIRE RATING

FLEXWELL®-HL has a two hour fire rating per UL 1369.

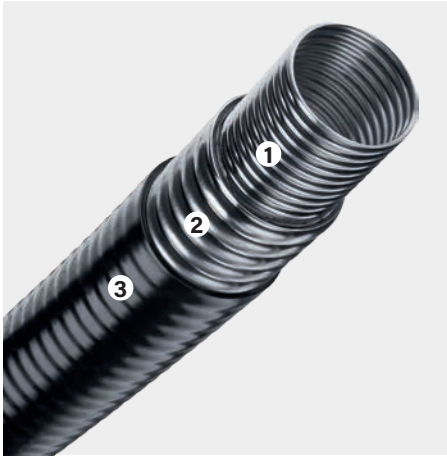


Closed mechanical end fitting – available in NPT or EZ-Fit

LEAK DETECTION

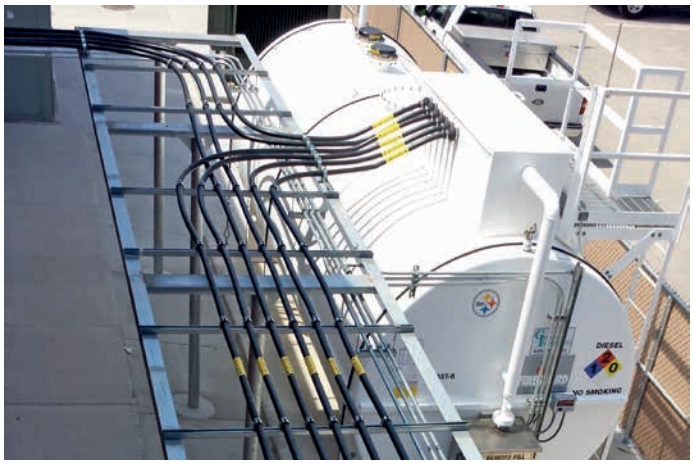
The interstital space between the primary and secondary pipes is designed for leak detection via either pressure or vacuum, depending on local regulations or operating requirements.

No requirement for constant leak monitoring per UL 1369 for BRUGG FLEXWELL®-HL.



FLEXWELL®-HL pipe

1 corrugated stainless steel primary pipe
2 corrugated stainless steel secondary pipe
3 polyethylene jacket



NEW: UL 1369 Above Ground Certified

Physical properties of FLEXWELL®-HL pipes

Model:		**HL 30/48**	**HL 48/71**	**HL 60/83**	**HL 98/134**
Material:	primary pipe: 316L (1.4404)	▪	▪	▪	▪
	secondary pipe: 316L (1.4404)	▪	▪	▪	▪
	corrosion protection: LDPE jacket	▪	▪	▪	▪
Operating pressure:	vacuum	suction operation	suction operation	suction operation	suction operation
Primary pipe (fuel etc.):	pressure	145 PSIG / 10 bar	145 PSIG / 10 bar	145 PSIG / 10 bar	145 PSIG / 10 bar
Nominal ID:		1" / ND 25	1 ½" / ND 40	2" / ND 50	3" / ND 80
Dimensions:	inside diameter (ID)	1.2" / 30 mm	1.9" / 48 mm	2.4" / 60 mm	3.9" / 98 mm
	outside diameter (OD)	1.9" / 48 mm	2.8" / 71 mm	3.3" / 83 mm	5.3" / 134 mm
	volume of primary pipe (gal/ft) / (liter/m)	0.06 / 0.8	0.16 / 2.0	0.24 / 3.0	0.67 / 8.4
Min. bending radius:	with bending machine	20" / 500 mm	26" / 650 mm	28" / 700 mm	40" / 1000 mm
Weight:	(lb/linear ft.) / (kg/linear m)	0.9 / 1.3	2.0 / 2.9	2.6 / 3.8	6.1 / 9.0
Permanent monitoring pressure secondary pipe:	vacuum	–8.7 PSI / –0.6 bar	–8.7 PSI / –0.6 bar	–8.7 PSI / –0.6 bar	–8.7 PSI / –0.6 bar
	pressure up to	145 PSIG / 10 bar	145 PSIG / 10 bar	145 PSIG / 10 bar	145 PSIG / 10 bar

All technical data subject to change.

PEI MEMBER

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